

Quality Inspection Report for Selective Laser Melting of ER70 Steel Marine Propeller

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Date of Inspection: October 26, 2024

Inspector: Maria Rodriguez, Senior Quality Engineer

Customer Reference: OceanTech Marine Propulsion Systems

Part Number: MP-ER70-SLM-001

Process: Selective Laser Melting (SLM) with Integrated 1 mm Machining Allowance

Applicable Standards: ASTM F3122-14 (Additive Manufacturing), ISO 484-1:2015 (Marine Propellers), Internal Quality Procedure QP-AM-03

Executive Summary

This report documents the quality inspection for the manufacturing of a single ER70 steel marine propeller produced via Selective Laser Melting (SLM). The inspection was conducted to verify conformance with specified material, dimensional, and process parameters. The propeller was built on an SLM Solutions 280 2.0 system using gas-atomized ER70 steel powder, followed by CNC finish machining to achieve final dimensions.

All critical-to-quality parameters were found to be within acceptable limits. The finished propeller mass was verified at 0.204 kg. Process consumables and energy usage were logged and aligned with expected operational ranges. The inspection concludes that the part conforms to all specified requirements and is approved for customer delivery.

1.0 Inspection Scope and Objectives

The scope of this inspection encompassed the complete manufacturing sequence for one (1) ER70 steel marine propeller, from raw material preparation through to final post-processing. The primary objectives were:

- To verify the quality and specification of incoming steel powder feedstock.
- To monitor and record key process parameters during the SLM build cycle.
- To inspect the final part geometry, surface finish, and mass.
- To document all material inputs, energy consumption, and waste outputs for traceability and process control.
- To ensure full compliance with the technical data sheet and customer purchase order specifications.

The inspection was performed in accordance with our internal quality management system, which is aligned with the referenced ASTM and ISO standards.

2.0 Process Description

The manufacturing process for the subject propeller consists of three primary stages:

1. **Powder Feedstock Production:** A certified ER70 steel billet was processed via inert gas atomization to produce the fine metal powder required for the SLM process. This sub-process consumes electricity, argon gas, and cooling water.
 2. **Selective Laser Melting (SLM):** The steel powder was deposited in thin layers within a build chamber filled with an argon atmosphere. A high-power laser selectively melted the powder according to the propeller’s digital CAD model, building the part layer-by-layer. This stage included the simultaneous fabrication of support structures, which were removed after the build.
 3. **Finish Machining:** The as-built propeller underwent CNC machining to remove the 1 mm allowance from all functional surfaces, achieving the final dimensional tolerances and surface finish specified by the customer. This operation consumed electricity and cutting fluid.
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3.0 Inspection Procedures and Methodology

A combination of measurement, monitoring, and visual inspection techniques was employed throughout the production process.

3.1 Material and Feedstock Inspection * Method: Certificate of Analysis (CoA) review for the steel billet; sieve analysis and Hall Flowmeter test for the produced powder. *** Standard:** Powder lot must conform to internal specification AM-MAT-ER70-01.

3.2 In-Process Monitoring * Method: Automated data logging from the SLM machine (power, gas flow, chamber atmosphere) and the CNC machine (power, fluid flow). All utilities are metered with calibrated instruments. *** Standard:** Process parameters must remain within the validated operating window defined in process work instruction WI-SLM-07.

3.3 Final Product Inspection * Dimensional Check: Performed using a coordinate measuring machine (CMM) with a probe accuracy of $\pm 1.5\text{ }\mu\text{m}$. Critical dimensions including pitch, chord length, and blade thickness were measured at 15 points per blade. *** Mass Measurement:** Weighed using a calibrated precision scale with an accuracy of $\pm 0.001\text{ kg}$. *** Visual Inspection:** Conducted under 10x magnification to check for surface defects such as cracks, porosity, and remnants of support structures.

3.4 Resource and Waste Tracking * All input materials (powder, gases, fluids) and output materials (part, waste) were weighed and recorded. Energy consumption was directly read from calibrated kWh meters on the respective equipment.

4.0 Inspection Results and Data

This section presents the detailed findings from the inspection activities. All data presented in the tables below pertains to the manufacturing of one (1) propeller and was recorded during the current production run.

4.1 Powder Feedstock Production Data

The gas atomization process was monitored, and the following consumables were used to produce the powder for this build.

Parameter	Quantity	Unit	Notes
Steel Billet (ER70)	0.414	kg	Raw material for powder production
Electricity Consumption	0.828	kWh	For atomizer operation
Argon Gas	2.58	kg	Inert atmosphere and atomizing gas
Cooling Water	0.116	kg	For system cooling
Output: ER70 Steel Powder	0.352	kg	Including powder for support structures

The electricity consumption for this batch was consistent with the typical energy intensity for this atomizer unit, which historically operates at approximately 2 kWh per kg of powder produced.

4.2 Selective Laser Melting (SLM) Build Data

The SLM build process was completed successfully. The total build time was 6.55 hours with an average machine power draw of 1.75 kW.

Parameter	Quantity	Unit	Notes
ER70 Steel Powder	0.352	kg	From Table 4.1
Electricity Consumption	11.49	kWh	For laser, scanner, motors, and auxiliaries
Compressed Air	3.56	m³	For powder handling and cleaning
Argon Gas	1.2	m³	For build chamber atmosphere

4.3 Finish Machining Data

The propeller was machined to its final dimensions on a 5-axis CNC milling center. The machining time was 4.133 hours with an average power consumption of 0.687 kW.

Parameter	Quantity	Unit	Notes
Electricity Consumption	2.84	kWh	For spindle, axes, and coolant pump
Cutting Fluid	1.94	kg	Water-soluble oil for cooling/lubrication

4.4 Final Product and Waste Output

The outputs from the manufacturing process are summarized below.

Parameter	Quantity	Unit	Notes
Marine Propeller (ER70 Steel)	0.204	kg	Final product, 1 unit
Machined Chips (Steel)	0.116	kg	Waste from 1 mm machining allowance

4.5 Dimensional and Visual Inspection Results

The CMM inspection confirmed that all critical dimensions of the propeller were within the specified tolerances of ± 0.05 mm. A summary of key measurements is provided below.

Dimension	Specified Value	Measured Value	Status
Propeller Diameter	152.40 mm	152.38 mm	Pass
Hub Diameter	45.00 mm	45.02 mm	Pass
Blade Pitch (at 0.7R)	120.50 mm	120.48 mm	Pass
Final Part Mass	0.204 kg	0.204 kg	Pass

Visual inspection under magnification revealed no critical defects. Surface finish on machined areas was measured at Ra 1.6 μm , meeting the specified requirement of Ra ≤ 3.2 μm .

5.0 Conformance Assessment

Based on the data collected and inspections performed, the following conformance statements are made:

- **Material Conformance:** The ER70 steel powder used for the build conforms to the chemical and physical requirements per its CoA and internal specification AM-MAT-ER70-01.
- **Process Conformance:** All recorded SLM and machining parameters (laser power, scan speed, hatch spacing, feed rates) were maintained within the approved operational ranges defined in work instructions WI-SLM-07 and WI-CNC-04.

- **Product Conformance:** The final marine propeller meets all dimensional, mass, and visual quality criteria as specified in drawing MP-ER70-SLM-001. The part is free from cracks, significant porosity, and other manufacturing defects that would affect its serviceability.

Overall Conformance Status: PASS

6.0 Conclusions and Recommendations

The manufacturing process for the ER70 steel marine propeller was executed in a controlled and repeatable manner. All quality inspection checkpoints were successfully passed, and the final product is confirmed to conform to all customer and internal specifications. The documented data on resource consumption and waste generation provides a complete record for this production lot.

Recommendations: * The collected process data should be archived as a benchmark for future production of identical or similar components. * No corrective or preventive actions are required at this time. The process is stable and capable.

This report certifies that the inspected propeller is approved for release and shipment to the customer.

Inspector Signature: _____

Maria Rodriguez

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